Google Cloud H2S

AI Builder Cup

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Veriqo

Know what to check before you buy.

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Problem Statement

Used-car buyers can't quickly tell a car's real condition or registration status before paying — so they overpay or get defrauded. Veriqo gives a fast, evidence-based pre-purchase check that shows exactly what to verify.

Brief about the idea

Veriqo turns a used-car buyer's photos + the registration number into one clear, evidence-tagged report in about a minute — before they pay or travel to see the car.

- **Gemini reads the photos** for visible issues (damage, rust, tyre wear, paint mismatch) with a confidence level.
- **A registration (RC) lookup** adds official facts: insurance validity, active loan, challans, blacklist status.
- **Deterministic rules combine both** into a transparent risk read and a list of what to get professionally inspected.
- **Every finding is labelled by source** — Observed, Record, Inferred, or Not checked — so nothing is a black box.

Veriqo is a red-flag triage — it tells you what to check. It never certifies a car as safe, and a photo is never treated as proof of a hidden mechanical fault.

Opportunities

- **How different is it?** Cars24 / Spinny inspect only the cars they sell; independent inspections are paid, physical and next-day. No self-serve AI photo check exists for ANY car or a private listing.

- **How does it solve the problem?** An instant first-pass BEFORE money or travel — it surfaces red flags and the exact questions to ask the seller, and flags what needs a mechanic.

- **USP — transparent provenance.** Every finding is tagged Observed (photo) / Record (RC) / Inferred (rules) / Not checked, and the risk score is just the sum of stated rules — auditable, not a mystery number.

Four provenance tags used across the product:

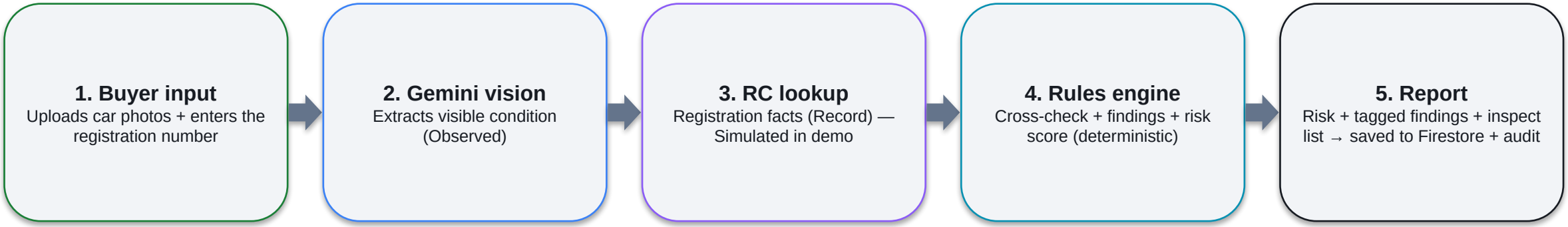
Observed | Record | Inferred | Not checked

List of features offered by the solution

- **Photo condition analysis (Gemini).** Visible damage, rust, tyre wear, paint mismatch — each with a confidence level.
- **Registration (RC) verification.** Make/model, insurance validity, active loan/hypothecation, challans, blacklist.
- **Evidence-tagged findings.** Observed / Record / Inferred / Not checked — the line is never blurred.
- **Transparent risk breakdown.** Score = sum of published rules across visible condition, record and uncertainty.
- **“Get inspected” action list.** Clear next steps and questions to ask before paying.
- **Shareable report + audit trail.** Every report is saved with an append-only log of how it was produced.

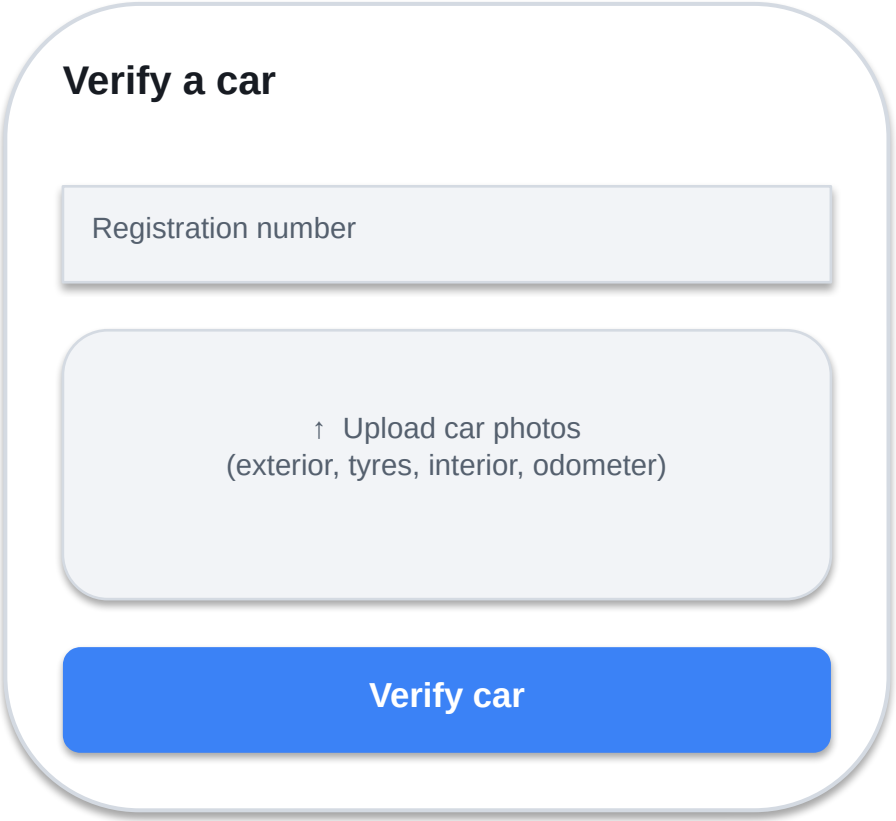
Honesty built in: when any source is mocked for the demo, the report shows a “SIMULATED DATA” banner and the RC record is labelled “not government-verified.”

Process flow diagram



One buyer action in, one transparent report out — each stage's output is traceable to its source.

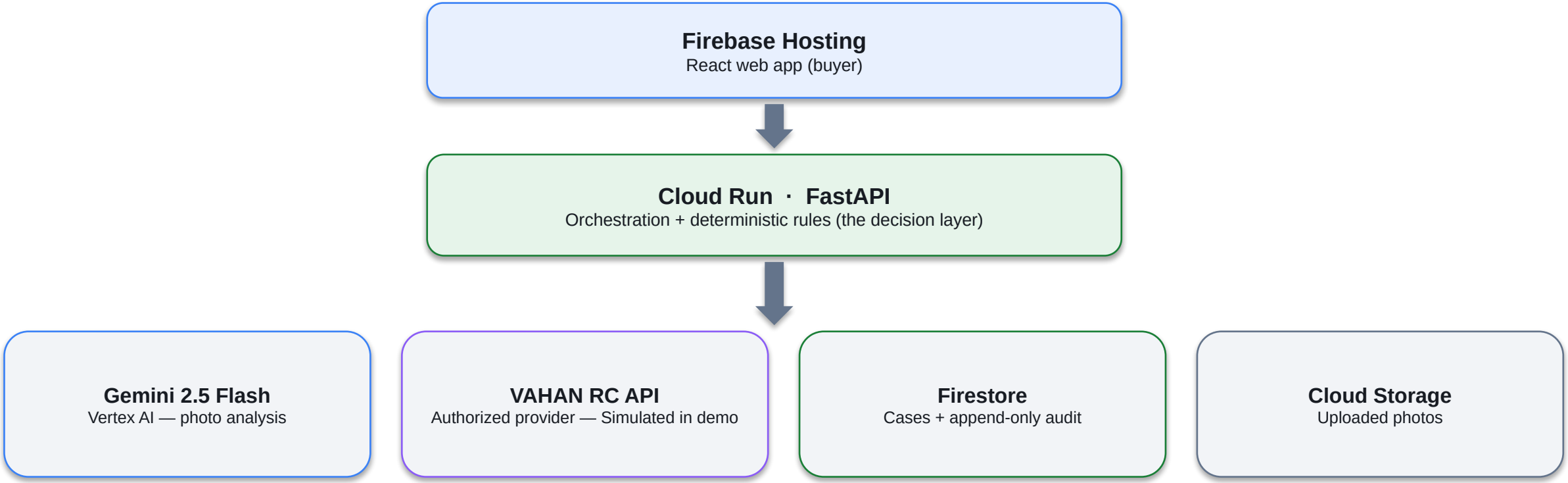
Wireframes / Mock diagrams of the proposed solution





Live prototype screenshots go on the “Snapshots” slide.

Architecture diagram of the proposed solution



Gemini perceives; Cloud Run decides with deterministic rules; Firestore keeps the auditable record. Only one external dependency (RC) is simulated until the live key is connected.

Technologies to be used in the solution

- **Gemini 2.5 Flash (Vertex AI).** Multimodal photo understanding with structured JSON output.
- **Cloud Run.** FastAPI backend: orchestration + deterministic scoring, autoscaling, kept warm.
- **Cloud Firestore.** Verification cases + append-only audit trail; shareable reports.
- **Firebase Hosting + Auth.** Serves the web app; rewrites /api to Cloud Run; optional sign-in.
- **Cloud Storage.** Stores uploaded car photos (evidence).
- **Authorized RC / VAHAN provider API.** Registration facts (Surepass / Signzy class) — simulated until key is live.
- **React + Vite + TypeScript.** Fast, responsive buyer UI.

Estimated implementation cost (optional)

Indicative, prototype-scale — built to stay within Google Cloud's free tiers; cost is dominated by the RC lookup.

Component	Basis	Indicative cost
Gemini 2.5 Flash	A few images per verification	Fraction of ₹ per report
RC / VAHAN lookup	Per registration lookup (provider-priced)	≈ ₹ a few per report
Cloud Run	Free tier; small cost if kept warm (min-instance)	≈ free – low ₹/month
Firestore / Storage / Hosting	Prototype volumes	Within free tier
Per-report (low volume)	**Sum, RC-dominated**	**≈ ₹ 5 – 15 (indicative)**

Figures are indicative estimates for a prototype, not a quoted price.

Snapshots of the prototype

[Insert screenshot]

Upload screen

[Insert screenshot]

Report with evidence tags

[Insert screenshot]

RC record + audit trail

ACTION: replace the three placeholders with real screenshots from the live deployed app.

Prototype Performance report / Benchmarking

How we measure Veriqo (honest method, no invented numbers):

- **Visible-issue recall.** On a labelled set of photo-sets with known visible issues, what fraction does Gemini flag?
- **False “all-clear” rate.** How often is a car with known visible issues reported with no flags? (Target: 0.)
- **End-to-end latency.** Time from submit to report — currently a few seconds per verification.
- **Transparency check.** Every finding carries a source tag and evidence; the score equals the sum of its rules (unit-tested).

[Insert your measured results here, e.g. “caught 9/10 known visible issues, 0 false all-clears across N cars; median latency ~X s.” Do not state numbers you have not actually measured.]

Additional Details / Future Development

- **Live RC integration.** Connect an authorized VAHAN provider to replace simulated registration data.
- **History signals.** Partner data for accident / service history and odometer-tampering checks (not available via API today).
- **Multilingual + voice.** Gemini-native local-language and spoken reports for wider reach.
- **Seller & dealer mode.** Shareable trust reports; an API for marketplaces and inspectors.
- **Adjacent products.** Same evidence-tagged engine applied to rental-listing verification (DwellCheck).

Provide links to your

- **GitHub Public Repository**

  [add public GitHub URL]

- **Demo Video Link (3 minutes)**

  [add video URL — must show the working flow]

- **Final Product Link (deployed)**

  [add live Firebase Hosting URL — must stay up during evaluation]

These three links are the evaluation gates. The deployed app must remain live and the video must clearly show the working flow — a broken link or a non-demonstrative video stops evaluation.

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Thank you

